

Score test

1. test statistic

$$Z^* = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

2. critical value

Two sided: $Z_{\frac{\alpha}{2}}$

Right sided: Z_{α}

Left sided: $-Z_{\alpha}$

3. decision

Two sided: If $|Z^*| > Z_{\frac{\alpha}{2}}$, reject H_0 .

Right sided: If $Z^* > Z_{\alpha}$, reject H_0 .

Left sided: If $Z^* < -Z_{\alpha}$, reject H_0 .

4. p-values

Two sided: $p - value = 2P(Z > |Z^*|)$

Right sided: $p - value = P(Z > Z^*)$

Left sided: $p - value = P(Z < Z^*)$

If $p - value < \alpha$, reject H_0 .

Wald CI

$$\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Agresti Coull CI

$$\tilde{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\tilde{p}(1-\tilde{p})}{n}}$$

Chi-squared test for independence

1. test statistic

$$\chi^{2*} = \sum \frac{(o - e)^2}{e}$$

2. critical value

$$df = (\#of\ rows - 1)(\#of\ columns - 1)$$

The critical value is $\chi^2_{\alpha, df}$

3. decision

If $\chi^{2*} > \chi^2_{\alpha, df}$, then reject H_0 .